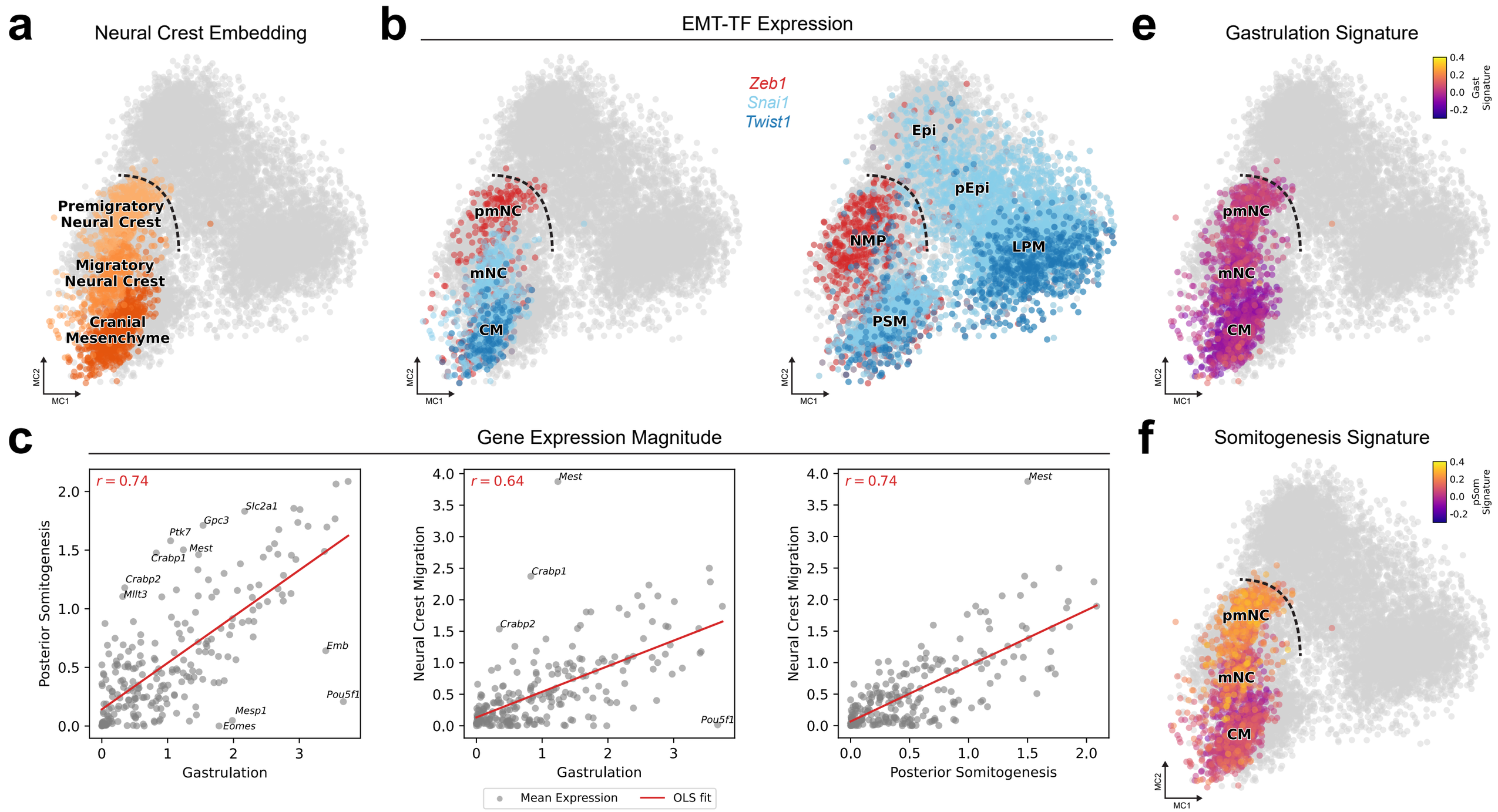

a
Neural Crest Embedding
Premigratory Neural Crest
Migratory Neural Crest
Cranial Mesenchyme
MC2
MC1
b
EMT-TF Expression
Zeb1
Snai1
Twist1
pmNC
mNC
CM
Epi
pEpi
NMP
LPM
PSM
e
Gastrulation Signature
Gast Signature
0.4
0.2
0.0
-0.2
f
Somitogenesis Signature
pSom Signature
c
Gene Expression Magnitude
r = 0.74
r = 0.64
r = 0.74
Posterior Somitogenesis
Neural Crest Migration
Gastrulation
Slc2a1
Gpc3
Ptk7
Mest
Crabp1
Crabp2
Mllt3
Emb
Pou5f1
Mesp1
Eomes
Mean Expression
OLS fit
d
Gene Expression Timing
r = 0.54
r = 0.49
r = 0.48
Early
Late
Meox1
Tcf15
Ebf3
Foxc2
Tmem200b
Six2
Arg1
Six1
Aldh1a2
Twist1
Ntn1
Itga9
Sertad4
Efnb2
Msx1
Tril
Nin
Lix1
Nkx1-2
Hes3
Ppfibp2
Crym
Sh2d3c
Bmpr1b
Aldh1a2
Efna3
T
Lfng
Wnt3a
Sp5
Ntn1
Dll1
Foxb1
Hand1
Fam43a
Mesp2
Nanog
Mean PseudoTime
g
Retinoic Acid Signaling
Somitogenesis
Meis2
Meis1
Crabp1
Rarg
Fabp5
Aldh1a2
Crabp2
Rbp1
Neural Crest
Rarg
Rarb
Rbp1
Meis2